

"At the End of the Day it's Your Best Friend. I Named Mine Sparky": Embodied Surveillance and the Use of Implanted Cardiac Devices

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Abstract

The use of implantable cardiac technologies reconfigures how device recipients relate to their bodies and the outside world. This article provides a situated extension of the concept of embodied surveillance and examines how people live and interact with implantable cardioverter defibrillators (ICDs) and pacemakers. We perform a thematic analysis of posts from ICD/pacemaker communities on Reddit. Focusing on the co-production of cardiac implants and bodies, we highlight how devices reshape sensations, routines, identities, and relations through continuous monitoring and occasional interventions. We present three overarching themes: uncertainties in embodiment, strategies and responses, and projections of roles and meanings, which together reveal how uncertainty and internal negotiation generate both new vulnerabilities and practices of resilience for device recipients. Cardiac implants function as intimate surveillance mechanisms that blur boundaries between care and control, contributing to surveillant assemblages in which agency is distributed across recipients, devices, and healthcare infrastructures. This study provides an illustration of the nuanced relationships between device recipients and implanted technologies.

CCS Concepts

• **Applied computing** → **Health care information systems**;
• **Human-centered computing** → *Human computer interaction (HCI)*; *User studies*; • **Computer systems organization** → *Embedded hardware*.

Keywords

Thematic Analysis, Reddit, ICD, Pacemaker, Cardiac Implants

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1 Introduction

The use of implantable cardiac devices such as internal defibrillators (ICDs) and pacemakers has drastically increased over the past decades, shifting from their original use in treating sudden cardiac arrest to serving as primary and secondary prevention treatment for a wide range of cardiac conditions [24, 31]. As invasive technologies surgically implanted within patients' bodies [27], ICDs and pacemakers reshape the everyday life and bodily experiences of device recipients.

ICDs and pacemakers are implantable devices composed of a generator and sensing leads. While ICDs typically detect and correct dangerously fast heart rhythms, pacemakers prevent the heart from beating too slowly. To deliver electrical impulses and shocks when needed, cardiac implants continuously track recipients' heart rate, with data stored and shared externally. In doing so, implants partake in datafication and act not only as medical care technologies, but also as systems that enact continuous surveillance. The human body is mediated and intimately monitored, simultaneously accommodating the device and reconfiguring the boundaries between self and technology, becoming a different version of itself [19, 35]. This study examines how ICD and pacemaker recipients describe and negotiate new conditions, vulnerabilities, and aspects of surveillance enacted by their cardiac implants in everyday life. We ask:

RQ1: How do individuals with ICDs and pacemakers describe their physical, psychological, and social experiences of living with these devices?

RQ2: How are practices of coping, self-regulation, and negotiation shaped by device monitoring and intervention capacities?

This qualitative study uses reflexive thematic analysis on a range of Reddit posts from subreddits related to ICDs/pacemakers and heart conditions. We use the concept of embodied surveillance, applying it to the accounts of individuals undergoing cardiac treatment with the use of ICDs and pacemakers. In so doing, we contribute to the conceptual understanding of contemporary surveillance practices in healthcare and affective experiences of medical devices.

This study adopts the notion of co-production [17] as a lens to examine how living with ICDs and pacemakers produces new enactments, practices, and vulnerabilities [21]. We thus understand cardiac implants and their recipients as mutually becoming, constituting each other's actions and vulnerabilities, and forming new bodies by becoming-affected [18].



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Bodies and technologies form assemblages in which each modifies and is modified by the other [2]. Surveillance systems in this study are also seen as assemblages, obscuring where the body ends and surveillance systems begin [12, 13]. We adopt assemblage theory when examining cardiac implant surveillance – the merging of care, control, and embodiment.

We understand surveillance as an embodied process, enacted through tracking and monitoring technologies that detect, classify, and respond to real-time changes in health and activity [11, 22]. The case of implanted devices complicates this picture. Unlike the Foucauldian panopticon, where internalization of the guard's gaze disciplines the subject, the surveillance of implantable cardiac devices is physically embedded within the body. Device recipients cannot fully know what actions may trigger a shock or pacing, their attempts to regain control become a part of the response to surveillance [10, 30]. We perceive cardiac implants to be a source of both care and control [20] and position them as technologies which perform embodied surveillance to describe forms of monitoring that are not merely intimate but physically merged with the body.

1.1 Literature Review

Within the biomedical discourse, attention has largely focused on the effectiveness and clinical outcomes of device use, prioritizing their capacity to extend physiological function [1, 4, 6, 16]. Studies in healthcare have focused on mortality, device functioning, and risk factors [20, 32], employing quantitative metrics such as device functionality, or quality-of-life scales. In contrast, science and technology studies (STS) has explored cardiac implants as sociotechnical and embodied phenomena, examining how these devices mediate lived experiences [7, 27, 28, 30]. In the HCI community, studies have focused on digital interventions, designing ethical health technology, and user interaction with monitoring technologies [8, 14, 33, 34]. Skov et al. [33] researched remote ICD monitoring, focusing on cardiac patients' perspectives on surveillance and privacy, claiming that individuals had very limited knowledge on how the system worked and what kind of data it collected. They describe a safety paradox, as participants felt less safe while monitored, despite actually being safer. There remains a need to examine the perspectives of device recipients and their interactions with medical technology and surveillance to recognize the complexity of personal embodied experiences.

2 Method

Reddit was chosen as a data source for this study because it contains thematic subreddits dedicated to ICD/pacemakers, with users sharing personal vulnerabilities and reflections [25, 29], which present as nuanced first-person perspectives. Anonymity of the platform allows the users to be open about their experiences outside of the clinical setting.

Reflexive thematic analysis [3, 5] was used to analyze publicly available posts from subreddits related to heart conditions and cardiac device therapy. We collected textual data from posts and comments, then inductively developed codes, subthemes, and themes, guided by our research questions. The analysis was performed on 110 posts and comments extracted from the r/PacemakerICD subreddit between January 2024 and October 2025, using purposive

sampling. Relevant posts were determined based on the author providing a public self-description or reflection on living with an ICD or a pacemaker, including sufficient detail. Additionally, comments that supported the discussions through personal narratives were included. After the initial codebook was created, deductive coding was used to apply it to 100 posts extracted from a broader range of subreddits (r/HypertrophicCM; r/Cardiology; r/askCardiology; r/Heartfailure). We organized and coded the data with NVivo 15, iteratively refining emergent codes and themes. Due to not being members of ICD/pacemaker communities, our analysis was shaped by our disciplinary training and personal standpoint, with codes reflexively constructed rather than assumed to emerge objectively from the data. We perceive the analyzed subreddits as sensitive spaces where users share intimate experiences of health conditions. To protect user anonymity, all usernames and potentially identifying details (e.g., dates, locations) were removed; excerpts were paraphrased to reduce searchability while preserving meaning. Data use followed contextual integrity by considering each subreddit's community norms and complying to Reddit's terms of service [9, 26].

3 Results

The analysis resulted in 396 codes with 10 subthemes, which we then grouped into three overarching themes: uncertainty in embodiment, strategies and responses to the device presence, and projection of roles and meanings onto the devices. Themes and subthemes are presented in Table 1. In the following sections, we briefly present the results of the analysis and examine how individuals undergoing cardiac device treatment navigate mutual constitution with an ICD/pacemaker, based on what they shared online.

3.1 Uncertainties in Embodiment

The Uncertainties in Embodiment theme refers to first-person descriptions of physical sensations and mental reactions in response to device presence. Posters described varying physical experiences, from not noticing the implant most of the time to having overwhelming and unfamiliar sensations. At times, the event that triggered a physical sensation was unknown, which led the poster to avoid specific actions or situations deemed as triggering: *"I got a shock yesterday. It was really hot, so I'm guessing the heat had something to do with it. I won't play again until I've seen my doctor in 2 weeks."* ICD shocks appeared as a major source of not only physical, but also psychological experiences. Device recipients described learning that their emotions and bodily responses may trigger the device, producing what they saw as a feedback loop: *"When I start feeling really anxious about it, my heartbeat speeds up, which only makes me more nervous. A vicious cycle."* Anxiety was perceived both as a potential cause of receiving a shock and a consequence of having been shocked. As a result, basic emotional responses gained a new dimension through becoming connected to the workings of the device.

These accounts illustrate how posters' physical and psychological reactions become a source of uncertainty, leading to unexpected or unpleasant events and alienating the individual from their body. The changed body becomes a source of sensitivity, with individuals paying attention and responding to new vulnerabilities. This

Table 1: Themes, Sub-Themes, and Representative Examples

Themes	Sub-Themes	Examples
Uncertainties in Embodiment	Physical Experiences	“I got a shock yesterday. It was really hot, so I’m guessing the heat had something to do with it. I won’t play again until I’ve seen my doctor in 2 weeks.”
	Psychological Experiences	“When I start feeling really anxious about it, my heartbeat speeds up, which only makes me more nervous. A vicious cycle.”
Strategies and Responses	Stress Management	“I try not to raise my voice, speak slowly, and slow my breathing to avoid any adrenaline spikes.”
	Therapy and Support	“Take things one day at a time, make sure you’ve got solid support around you, and get therapy! My EP said it’s PTSD and that I should go to psychotherapy to work through it.”
	Self-tracking	“I’ve got a cheap watch that I use as a heart rate monitor, and I’m checking it all the time.”
	Modifying Lifestyle and Identity	“I’m a big gamer, and I really like souls-like games (the tough, stressful ones). I noticed my heart rate shoots up when things get intense, so my doctor told me to stop playing games that stress me out.” “While running one day, my heart acted up and delivered another shock—the first since my original episode. I panicked, went to the hospital, and they said I’d likely overdone it. After that, I quit running, afraid it might happen again.”
	Negotiating Conditions	“I keep a magnet with me during riskier activities so I can shut off the ICD if I were ever seriously wounded.” “I had a checkup a few weeks back and asked why my heart rate was so low. They told me it was set according to my lifestyle, but I got my pacemaker ten years ago, and my lifestyle’s different now.”
Projections of Roles and Meanings	Psychological and Linguistic Expressions	“The fun part is that I can tell my kids I’m a cyborg!” “I’m 34 now, and I’ve never had to use the shocky-shock part of it.”
	Safety, Gratefulness and Fortune	“Obviously it’s better than the alternative and we should frame it with gratitude.”
	Ambivalence and Disappointment	“I know my ICD literally saves my life, but it’s tough not being able to do the things I enjoy.” “Seems like my device saved me from something that could’ve been life-threatening. I’m glad it worked, but now I’m really scared about my health.”

exemplifies co-production, both the device and the individual affecting each other through psychological and physical enactments. In this case, embodied surveillance extends beyond monitoring by being reactive — the enactments of the device are followed by bodily responses and psychological reactions, which are once again sensed by the device. This highlights the fluid nature of embodied surveillance, as opposed to static monitoring.

3.2 Strategies and Responses

The Strategies and Responses theme represents a range of responses to device presence described by individuals. These practices represent reactions to the common vulnerabilities of device treatment.

Posters described feeling that their own emotions can affect the device. Thus, their attention to emotional reactions increased, and staying calm became a precaution measure. Individuals mentioned strategies of emotional self-regulation: *“I try not to raise my voice, speak slowly, and slow my breathing to avoid any adrenaline spikes.”* Such practices exemplify how embodied surveillance influences sensitivity to emotional triggers and bleeds into self-governance: a cardiac device becomes a factor in how individuals speak, breathe, and conduct themselves. Posters occasionally mentioned having a support group, doing psychotherapy, meditating, and finding comfort in spiritual practices to overcome cardiac anxiety from being hyper-aware of possible shocks. Some described relying on various types of tracking or external monitoring, trying to pick up on signs of possible heart instability and predict when to expect shocks. By sharing their data to external actors, some individuals reported feeling less vulnerable in the face of possible complications.

Another response to vulnerabilities of device treatment can be seen in changes of lifestyle and identity. The modifications described in the posts spanned multiple life aspects: physical exercise, hobbies, nutrition, public engagement, and workplaces. For example, posters described hobbies such as gaming, music, and sports as central to their identity, yet risky after implantation. Some stopped “hard and stressful games,” others adjusted how they hold a guitar or limited physical activity. These negotiations show how the device subtly redraws what forms of leisure and self expression become allowed: *“I’m a big gamer, and I really like souls-like games (the tough, stressful ones). I noticed my heart rate shoots up when things get intense, so my doctor told me to stop playing games that stress me out.”*

Discussions also brought up shifting work conditions and job opportunities. Particularly, the workplace limitations appeared to play a bigger role in the blue-collar sector, as many of the jobs involve dealing with electrical appliances and performing physical labor. Welding and rail industry jobs were brought up as examples of high voltage environments that are not cardiac device friendly: *“I decided to get the ICD, which basically meant my career was over.”* To accommodate the changed body to the working conditions, one poster mentioned using a magnet adaptively in work environments or social scenarios: *“I keep a magnet with me during riskier activities so I can shut off the ICD if I were ever seriously wounded.”* Additionally, posts mentioned lowering or increasing the sensitivity of the device, adjusting pacing ranges and defibrillation thresholds or disabling the rate response with the help of healthcare providers. Planning, tracking, packing the hospital bags or wearing a safety

bracelet with emergency contacts came up as strategies for coping with unpredictable cardiac events. Describing their social interactions, several posters brought up not mentioning their device recipient status while filling out forms or communicating with security. Others mentioned having to specify to the personnel that certain security protocols can’t be applied to their bodies because of the potential effect they have on the implant. Those accounts exemplify attempts at changing the functioning of devices or working around emerged limitations to negotiate the new conditions of life with the cardiac implant.

The described strategies represent the ways in which embodied surveillance reconfigures the daily lives of individuals, impacting not only their sensations, but also behaviors and routines. We highlight the active negotiations between individuals and their devices, their mutual becoming unfolding through daily interactions in various contexts.

3.3 Projections of Roles and Meanings

We link the influence of implanted devices on individuals’ daily practices to them developing underlying attitudes towards the implant, its functioning and its role within one’s life. The Projections of Roles and Meanings theme represents a range of attitudes towards cardiac implants, showing expectations and values projected onto the technology.

The use of humor and metaphoric language often came up as a response to device presence. Several posts mentioned using humor to focus on the benefits of living with an implant and project positive views: *“The fun part is that I can tell my kids I’m a cyborg!”* Another example appeared in the use of euphemisms to describe events that may be traumatic or uncomfortable, framing them as playful (“getting zapped”; “riding the lightning”; “electric storm”; “blasting”). In some of the posts, embodied technology was personified (“best friend”; “guardian angel”; “little paramedic”). One poster wrote: *“At the end of the day it’s your best friend. I named mine Sparky.”* We connect personification to attributing values such as friendliness and competence to the device, building trust towards the implant and mitigating friction. Posters often viewed the device as a positive presence, referring to it as a “safety net” or “safety blanket.” Some described a sense of comfort from being watched over by the implant, emphasizing that surveillance offered reassurance and protection from risk. Another type of positive attitude was represented through expressing gratitude for receiving a device. Posters described feeling thankful for surviving near-death situations, framing it as receiving a second chance at life: *“Obviously it’s better than the alternative and we should frame it with gratitude.”* These expressions may reflect emotional engagement with the implant, highlighting how recipients not only value the device for its medical function but also project symbolic meaning of hope and positive change onto it. Alternatively, some individuals felt ambivalent, with expressions ranging from positively reassuring (“It will get better with time”) to unwillingly accepting the arrangement (“There is nothing I can do about it”). In these accounts, the device was viewed as a tool that performs its intended function, yet also serves as a reminder of vulnerability. Sometimes, vulnerabilities resulted in disappointment, posters claiming that maintaining usual routines became impossible after receiving an implant, seeing it as

a limiting factor: “I know my ICD literally saves my life, but it’s tough not being able to do the things I enjoy.” We interpret such accounts as frustration directed toward the implant, which functions as a mediator between human perception and internal physiological conditions.

These examples illustrate how embodied surveillance transforms from clinical to personalized and intimate. Individuals perceive it as nuanced — playful, hopeful, and limiting, rather than simply positive or negative. The accounts further highlight the role of personal interpretation in implanted device surveillance and its effect on subjectivity.

4 Discussion

In this study, we explored embodied surveillance in the lived experiences of ICD and pacemaker recipients through public accounts on Reddit. The posts fell broadly within three themes: uncertainties in embodiment, strategies and responses to the device presence, and projections of roles and meanings onto cardiac devices. Through our analysis, we posit ICDs and pacemakers as embodied surveillance infrastructures rather than purely clinical devices. We illustrate how individuals face vulnerabilities and continuously form co-produced bodies, responding to the duality of embodied surveillance as both caring and controlling.

As seen in the findings, individuals describe feeling their implants in multiple ways — by getting new bodily sensations, becoming more aware of their reactions, and modifying their habits, hobbies and lifestyles. We suggest these embodied reactions are a consequence of surveillance. Monitoring technologies instigate a surveillance relationship between a person and their device, as well as mediation relations [15, 35]. Simultaneously, device recipients modify and are modified by their experiences with embodied technology, with both the body and technology developing new sensitivities that lead to new practices as well as anxieties. Within this, we recognize surveillance performed by implants as not only monitoring but also as part of an intimate and ongoing becoming of body with technology.

Using the term “embodied surveillance” in the context of implanted technology, we extend the previously developed concept of embodied surveillance as one in which the observed and the observer are known to one another [23]. In the use of cardiac implants, surveillance occurs in multiple forms, unfolding within the body and extending to external infrastructures, forming assemblages. Embodied surveillance refers to the intimate mechanism of surveillance enacted within the body by devices that continuously track and intervene in physiological functions. This process reshapes bodily awareness, impacts physical and psychological sensitivity, and produces new vulnerabilities. Moreover, the human body and the implanted technology become part of the surveillant assemblage [12, 13] through the processes of sensitizing, self-regulation, and mutual becoming with the device.

The findings show how embodied devices affect human actions and perceptions of illness, physical limitations, and well-being. We display the negotiation of frictions between the device and its recipient performed through material interactions, communication and personal reflections. The device’s presence becomes a source of medical care for the body, and a source of concern. Becoming a

part of an assemblage, the body is multiplied and sensitized, new practices emerging as attempts to care for this arrangement. Values, expectations, and roles are projected onto the device, as seen in expressed attitudes, attempts at personification, and humor. These complexities are neither exclusively produced by the implant nor the individual; they arise from their intertwined relationship, illustrating the affective side of embodied surveillance. This study contributes to understanding the workings of co-production and surveillance, describing how they play out through interactions, perceptions and sensitivities. We illustrate how bodies and cardiac devices co-evolve through continuous interactions, revealing the unexpected ways individuals engage with and are reshaped by implanted technology and surveillance, resonating with prior research on patient agency and technological embodiment.

5 Limitations

The results were obtained from a small data sample, limited in diversity and scope. The choice of Reddit as a data source involves participants who are digitally literate and English-speaking, which does not represent all ICD/pacemaker recipients and negates cultural contexts. The analysis was performed on naturally occurring data, which excludes the possibility of clarification or follow-up. Findings illustrate possible experiences and practices rather than a generalized representation. The results of this study will be supported by future work that employs interviews with more diverse participants, longitudinal studies, or comparative work across platforms and devices.

6 Conclusion

The results of this research emphasize the enactment of surveillance as an intimate process that influences experiences of embodiment, identity, and agency, as opposed to it being strictly an element of social control. They also highlight the role of surveillance in co-evolving of bodies and technologies and experiencing a changed body as a result of technological intrusion. We apply the theoretical concept of embodied surveillance in implantable devices. Future research should build on this analysis by exploring the evolving patient-device dynamics in the broader scope of implanted and wearable technologies.

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